Set-2

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EVEN END SEMESTER EXAMINATION JUNE – 2025

Name of the course: B. Tech
Name of Paper: Automation in Production
Time: 3 Hours

Semester: 4th
Paper code: TME410
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1. (2X10=20 Marks)

a. Compare and contrast the direct and indirect costs in a manufacturing system. CO2

b. Compare different voltage divider configurations (resistor-resistor, resistor-photoresistor, resistor-switch, resistor-LED) and analyze their performance when used as analog input sensors in Arduino. CO4

c. Assume that the frequency of line stop occurrences in Problem 5.1 is due to random mechanical andelectrical failures on the line. Suppose, in addition to the foregoing reasons for line stops, that theworkstation tools must be reset and/or changed at regular intervals. It takes an average of 4 min to inspectand adjust or change the tools at all eight stations. This procedure is performed every 200 cycles.Recompute Tp, Rp, E, and D with these additional data for the indexing machine. CO2

Q2. (2X10=20 Marks)

a. What are Automated Storage/Retrieval Systems (AS/RS)? CO3

b. Discuss the various basic industrial sensors such as: CO5

i) IR Analog Sensor
ii) IR Digital Sensor
iii) Color IR (TSOP) Sensor
iv) Provide working principles and typical industrial applications for each

c. Explain the various problems encountered in interfacing handling and storage systems with manufacturing units. CO3

Q3. (2X10=20 Marks)

a. Differentiate between belt conveyors and roller conveyors based on working principle and application. CO3

b. List and explain the design and fabrication considerations for developing a new automated flow line. CO2

c. a) What is the objective of line balancing problem? CO2

b) Differentiate inspection and testing.

Q4. (2X10=20 Marks)

a. Define storage system performance. Why is storage system throughput important? CO3

b. What is a voltage divider circuit? CO4

Explain its working using three cases:

(a) Resistor/Resistor

(b) Resistor/Photoresistor

(c) Resistor/LED

Draw all three circuits and calculate output voltages for each using assumed component values.

c. Suppose that a 10-station transfer machine is under consideration to produce a component used in a pump. 'The item is currently produced by more conventional means, but demand for the item cannot be met. The manufacturing engineering department has estimated that the ideal cycle time will be Tc = 1.0 min. From similar transfer lines, it is estimated that breakdowns of all types will occur with a frequency, F = 0.10breakdown/cycle, and that the average downtime per line stop will be 6.0 min. The scrap rate for the current conventional processing method is 5% and this is considered a good estimate for the transfer line. The starting casting for the component costs $1.50 each and it will cost $60.00/h or $1.00/min to operate the transfer line. Cutting tools are estimated to cost $0.15/work part. CO2

(a) Production rate.

(b) Number of hours required to meet a demand of 1500 units/week.

(c) Line efficiency.

(d) Cost per unit produced.

Q5. (2X10=20 Marks)

a. Describe the selection criteria for choosing a sensor for an industrial application. CO1

b. Describe the working principles of capacitors, resistors, inductors, and diodes. CO4

Provide their circuit symbols, characteristics, and at least one application for each component.

c. Perform an analytical comparison between a fully automated flow line and a partially automated flow line. CO2